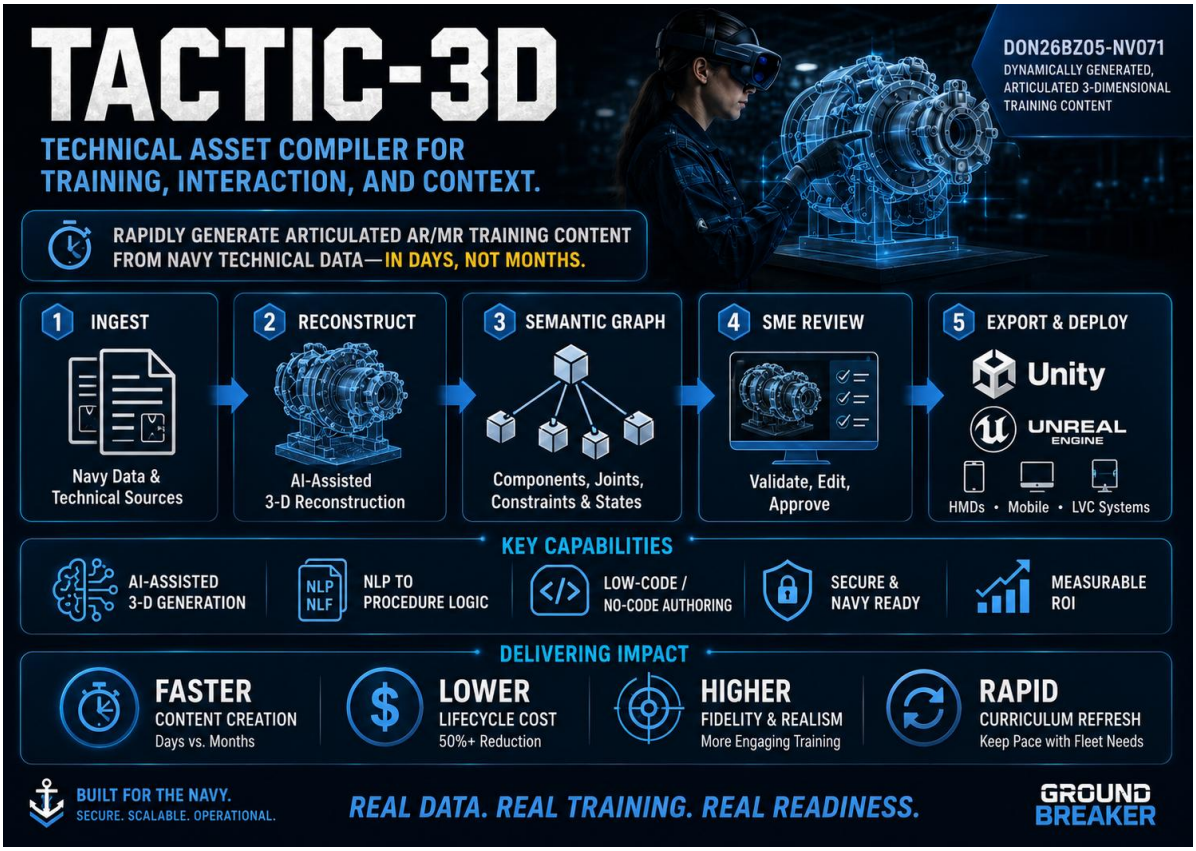
TACTIC-3D — Technical Asset Compiler for Training, Interaction, and Context

1. Identification and Significance of the Problem or Opportunity

The Navy seeks a secure, user-friendly toolkit that rapidly converts heterogeneous structured and unstructured fleet data into photorealistic, articulated, interactive 3-D training content; gives Subject Matter Experts (SMEs) low/no-code control; and exports to AR/MR and Live-Virtual-Constructive (LVC) environments. The topic identifies the central gap: recent generative 3-D methods can accelerate scene creation, but current solutions often lack the dynamic articulation, physics, procedural logic, SME oversight, multi-user integration, and standards-compliant export required for hands-on procedural training [1]. Traditional production therefore remains dominated by manual modeling, rigging, scripting, integration, and verification, slowing curriculum refresh and raising lifecycle cost. Groundbreaker proposes TACTIC-3D, a human-supervised procedural digital-twin compiler. TACTIC-3D will ingest engineering, visual, and technical-procedure evidence; construct a Semantic Asset Graph (SAG) that binds components to geometry and source evidence; infer candidate joints, constraints, and interaction affordances; compile technical instructions into deterministic procedural state logic; route uncertainty to a no-code SME validation workflow; and publish a validated, engine-neutral training package through Unity/Unreal adapters. Phase I will not claim autonomous generation of safety-critical training. It will determine experimentally under what source-data conditions AI assistance can create a technically correct interactive representation with materially less human authoring effort.

1.1 Innovation

TACTIC-3D separates appearance from behavior. NeRF and 3-D Gaussian Splatting (3DGS) are powerful for view synthesis and photorealistic rendering, but neither representation inherently supplies the engineering part hierarchy, rigid-body joints, collision volumes, tools, hazards, procedure preconditions, faults, or state transitions required by a maintenance trainer [2,3]. TACTIC-3D treats reconstruction/foundation models as replaceable components and preserves the durable behavior in two prime-owned canonical representations: the SAG and a Procedure Intermediate Representation (Procedure IR). A generated access cover is therefore not merely a mesh; the system represents what secures it, when it may be removed, what becomes accessible afterward, and which safety state must hold before interaction.



1.2 Phase I Demonstration

Phase I will instrument one Navy-representative articulated maintenance asset—nominally a centrifugal pump or comparable electromechanical assembly—and one primary maintenance procedure plus a fault/variant. The source pack will include available CAD/assembly data, schematics or exploded views, an authoritative maintenance procedure, smartphone imagery/video, and representative maintenance/log data. TACTIC-3D will generate a componentized interactive asset, compile the procedure, permit no-code SME correction, execute the validated scenario in a primary engine, and demonstrate critical behavioral conformance in a second engine. Results will be measured against an independently prepared SME gold standard and a matched manual-authoring baseline, directly addressing the topic requirement to track time to generate, test, and integrate [1].

2. Phase I Technical Objectives and Research Hypotheses

Phase I is organized as a falsifiable research program. The key question is not whether AI can generate an impressive visual demo, but whether the coupled pipeline produces correct training behavior with a quantifiable reduction in human effort.

Objective	Phase I hypothesis/target	Measures
O1 Multimodal semantics	Fusion of CAD/visual/document evidence achieves ≥0.85 F1 on training-relevant component identity and exceeds the strongest single-modality condition by ≥0.05 F1.	F1; entity-link accuracy; ablations.
O2 Interaction-ready asset	Hybrid geometry represents ≥90% of SME-designated components as usable scene nodes/collision proxies at correct scale.	Coverage; scale; collider validity; time.
O3 Articulation	Correct joint class ≥85%; ≥70% first-pass SME acceptance; ≥95% correct after one correction cycle.	Joint/axis/limit error; accept/edit/reject.
O4 Procedure compilation	Recover ≥95% of procedure steps; top-1 component mapping ≥90%; 100% provenance for generated assertions.	Step recall; mapping; provenance.
O5 Productivity/portability	Reduce human labor to accepted scenario ≥50% vs. manual baseline while critical state tests pass in both adapters.	Labor; SME edits; conformance; critical errors.

2.1 Experimental Comparators and Source Ablation

The evaluation uses three comparator classes. A competent XR developer plus SME will manually author the same defined scenario to the same acceptance criteria. Single-modality conditions (CAD-only, visual-only, document-only) will determine whether multimodal fusion adds measurable value. Algorithmic baselines will compare classical photogrammetry with neural appearance methods, deterministic/template articulation with multimodal hypotheses, and rules-only parsing with source-grounded LLM-assisted extraction. Minimal-source experiments will evaluate: full evidence; CAD+manual; photos+manual; video+manual; visual-only; and manual/schematic-only. This design tells the Navy what minimum evidence is needed for a desired confidence level rather than assuming every data source is always available.

3. Technical Approach and Architecture

TACTIC-3D uses a canonical, engine-independent architecture separating four concerns that are often collapsed: physical geometry, visual appearance, semantic identity, and procedural behavior. Stable interfaces allow individual AI/reconstruction models to be replaced as the field advances. The pipeline is: (1) secure evidence ingestion; (2) CAD/geometry and technical-data parsing; (3) cross-modal entity resolution; (4) SAG construction; (5) articulation/physics inference and Procedure IR compilation; (6) deterministic validation and no-code SME review; and (7) a signed TACTIC package consumed by Unity, Unreal, and future Navy/LVC adapters.

3.1 Secure Multimodal Evidence Ingestion

Every input becomes an immutable EvidenceItem with a stable ID, cryptographic hash, source type, acquisition metadata, project/handling label, parser/model version, and source location. CAD ingestion preserves assembly hierarchy, part identifiers, transforms, and engineering metadata instead of flattening the asset. Images/video retain capture order and available camera metadata. Technical publications preserve page/section/table/figure locations so derived assertions can be traced to evidence. Logs and maintenance records are normalized without discarding original records. When S1000D is available, TACTIC-3D will consume its structured procedural, fault, illustrated-parts, and related data modules rather than apply unconstrained NLP to already-structured technical data [8].

3.2 Semantic Asset Graph, Provenance, and Confidence

The SAG is the center of TACTIC-3D foreground R&D. Phase I will implement a versioned JSON/JSON-LD-compatible canonical model with Asset/Assembly/Component, GeometryRef, Joint, Affordance, Tool/Consumable, Procedure/Step, State, Constraint/Hazard, Evidence, Validation, and ExportBinding nodes. Relationships include partOf, connectedTo, articulatedBy, affords, actsOn, requiresTool, requiresState, producesState, prohibits, derivedFrom, and validatedBy. Provenance borrows W3C PROV-O concepts of Entity, Activity, Agent, derivation, and attribution [9]. Every AI assertion retains evidence IDs, model/parser version, inference timestamp, confidence, and SME decision. Confidence controls workflow: high-confidence suggestions are "accept-ready" only after empirical calibration; safety-critical assertions always require explicit SME approval.

3.3 Hybrid Geometry and Reconstruction

When assembly CAD exists, it is the primary source of metric geometry and hierarchy; TACTIC-3D simplifies it into interaction meshes/colliders and uses imagery to align the model with the as-maintained asset. When CAD is unavailable, Structure-from-Motion/Multi-View Stereo is the primary mesh-generating baseline. NeRF and 3DGS are evaluated as appearance/reconstruction technologies because they provide strong novel-view fidelity [2,3], but the architecture does not assume radiance representations are production collision geometry. A hybrid asset may therefore contain CAD/low-poly/procedural interaction geometry, colliders and joint frames, plus a photorealistic surface/splat representation mapped to the same semantic component. A Month-3 decision gate selects the canonical Phase I geometry path using component coverage, scale accuracy, collider usability, fidelity, generation time, and SME correction burden.

3.4 Component Identity and Articulation

Cross-modal entity resolution reconciles CAD/BOM names, schematic callouts, manual terminology, visual regions, and procedure references into canonical component IDs. Ambiguous aliases are presented as ranked candidates rather than hallucinated matches. Articulation evidence is fused in descending reliability: CAD assembly constraints; explicit manual/schematic statements; observed motion from two-state imagery/video; geometry/mechanical priors; and multimodal model hypotheses. PARIS demonstrates simultaneous part reconstruction and motion estimation from two multi-view articulation states [4], while Articulate-Anything demonstrates vision-language-driven articulated modeling and iterative correction [10]. TACTIC-3D uses these as feasibility evidence/baselines, not as a single mandatory dependency. Each ArticulationHypothesis stores parent/child, joint class, frame/axis, limits, default state, collision relation, affordance, evidence, confidence, and SME validation.

3.5 Procedure Compiler and Deterministic Validation

A language model will never convert a maintenance manual directly into executable Unity/Unreal lesson code. It produces a constrained ProcedureStep draft containing sourceEvidence, action, targetComponent, tool/consumable, preconditions, safetyConstraints, parameters, expectedPostconditions, and branches. Entity resolution binds textual targets to SAG IDs. A deterministic validator verifies that components exist, required safety gates are sourced, preconditions are consistent, and the transition graph is executable. The compiler lowers accepted steps into Procedure IR/state transitions. "Remove the access cover after removing retaining fasteners" becomes a rule that Remove(Cover) is enabled only when designated fasteners are Removed and the required safety state holds; successful execution changes Cover to Removed and exposes downstream components. This supports deterministic testing, fault injection, telemetry, and engine portability while keeping the foundation model out of the runtime safety loop.

3.6 Low/No-Code SME Workflow

The Phase I editor provides synchronized source-evidence, 3-D, and semantic/procedure views. An SME can accept/reject identities, rename aliases, associate geometry, adjust a joint axis/limit, map/reorder/branch steps, define a prohibited action, inspect provenance, and preview the scenario without C#, Blueprint, or Python. Every correction is versioned and attributable. This directly addresses the topic requirement for human-on-the-loop validation and rapid instructor/curriculum-developer authoring [1].

3.7 Runtime Export, XR/LVC, and Cybersecurity

A small runtime adapter in each engine consumes the canonical package instead of generating unique lesson code. The package contains portable runtime geometry (for example GLB), optional OpenUSD scene/physics data, SAG, Procedure IR, provenance, and conformance tests. glTF is maintained by Khronos as a runtime 3-D asset format [6]; OpenUSD physics schemas explicitly represent rigid bodies and joints [5]. Unity AR Foundation supports multi-platform AR [7], Unreal provides extensible Interchange/USD workflows [11], and OpenXR 1.1 provides a cross-platform XR API target [12]. Phase I will demonstrate one primary engine and critical conformance in the second; Phase II expands device/LVC integration.

The topic projects CMMC Level 2 (Self) [1]. TACTIC-3D will be CUI-boundary aware from Phase I: least privilege, project/role access, immutable audit, handling labels on evidence/derivatives, SBOM and model/dependency inventory, signed/versioned packages, and no requirement to send Navy technical data to a public AI service. A model abstraction layer supports approved local/private-cloud inference. Runtime adapters consume validated declarative packages rather than arbitrary model-generated code. Current DoW guidance ties broad protection of CUI at CMMC Level 2 to NIST SP 800-171 requirements [13].

3.8 Navy Training Process Alignment

The topic cites NAVEDTRA 130B [1]. NETC now lists the NAVEDTRA M-142 Navy Training Process series dated June 18, 2025, covering requirements, course development/modernization, fielding/testing, assessment, and sustainment [14]. TACTIC-3D will preserve task, condition, standard, source, version, approval, test, and sustainment metadata so generated modules fit the current training lifecycle rather than exist as isolated XR assets.

4. Phase I Experimental Methodology and Evaluation

4.1 Test Case and Gold Standard

The integrated test uses one representative articulated maintenance asset with enough relevant mechanics to exercise valves/handles, covers, fasteners, switches, caps, and access states. The target source pack is assembly CAD if available; BOM/parts data; schematic/exploded view; one authoritative maintenance procedure; approximately 60–120 multi-view photographs or controlled walk-around video; imagery/video of at least two articulation states when practical; and representative maintenance/log data. These are test-design assumptions, not solicitation requirements. Before TACTIC-3D runs, an independent SME labels the component hierarchy, aliases, geometry mappings, joints/axes/limits, collision-critical surfaces, procedure steps, step-to-component mappings, tools, pre/postconditions, safety constraints, fault opportunities, and source evidence.

4.2 Instrumented Time Study

The manual and TACTIC-assisted workflows target identical acceptance criteria. Instrumentation separately records active human authoring, SME review, correction, AI/GPU processing, asynchronous wait, runtime integration, test/debug, and total elapsed time. The primary ROI result is human labor to an accepted interactive scenario; elapsed generation, test, and integration times are reported separately as required by the topic [1]. Failures are tagged as ingestion, geometry, segmentation, entity-resolution, articulation, procedure extraction, logic, provenance, usability, or adapter errors.

4.3 Phase I Scorecard

Metric	Research target
Component ID F1	≥0.85
Component↔geometry association	≥90%
Joint-class accuracy	≥85%
Articulation first-pass SME acceptance	≥70%
Articulation correct after one SME edit	≥95%
Procedure-step recall	≥95%
Step→component top-1 / top-3	≥90% / ≥97%
Provenance / safety-critical SME approval	100% / 100%
Procedure conformance	≥95% overall; 100% critical
Human labor	≤16 hr and ≥50% below manual baseline
Substantive SME edit fraction	≤30%
Cross-engine conformance	≥95% overall; 100% critical
High-confidence suggestion precision	≥95%

Targets are research objectives, not claimed existing performance. Point estimates will be accompanied by 95% bootstrap confidence intervals where sample sizes permit. The final demo uses a frozen source package and repeats: ingest → semantic inventory → geometry/articulation → procedure compilation → SME correction → deterministic validation → export → valid and invalid/fault execution → secondary-engine conformance → measured ROI. Phase I succeeds only if the complete scorecard, including failures, is reported.

5. Phase I Base Statement of Work — Six Months

The Phase I Base is exactly six months under current DON Release 5 instructions; Groundbreaker will retain well over the required two-thirds of the work. Specialized subcontracting is limited to XR/runtime integration and focused 3-D reconstruction/articulation advisory support [15].

Task 1 — Research protocol, requirements, and security boundary (Month 1). Trace NV071 requirements; freeze the pump use case and source conditions; create the independent gold-standard protocol; define H1–H5 metrics, manual baseline, source ablations, security boundary, and model/dependency/license register. Exit: all metrics are computable and acceptance criteria are frozen.

Task 2 — Multimodal EvidencePack (Months 1–2). Implement CAD, image/video, document, schematic/log adapters; immutable evidence metadata; component alias normalization; and S1000D pathway. Exit: ≥95% of the test source files are ingested without manual restructuring and the first semantic component inventory is generated.

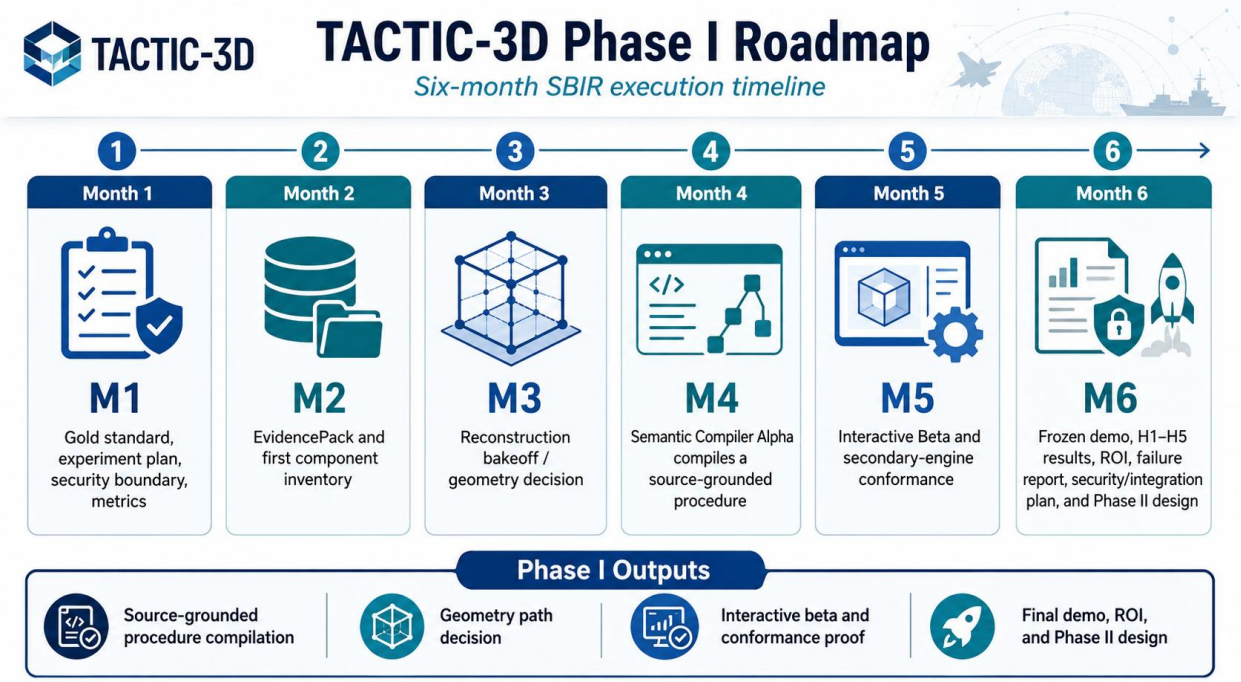
Task 3 — Geometry, SAG, and articulation bakeoff (Months 1–4). Compare CAD/hybrid, SfM/MVS, and optional neural appearance; implement SAG and component↔geometry resolution; generate joint/collider hypotheses; run the Month-3 geometry decision gate. Exit: selected geometry path, SAG alpha, and measured component/articulation results.

Task 4 — Procedure compiler and SME editor (Months 2–5). Implement source-grounded extraction, entity linking, Procedure IR, deterministic validation, provenance/confidence, and no-code review. Exit: one maintenance procedure compiles end-to-end and an SME can correct component/joint/procedure information without code.

Task 5 — Runtime adapters and conformance (Months 4–6). Implement primary Unity or Unreal runtime, secondary-engine conformance/import proof, interaction bindings, telemetry, fault path, package versioning/signing. Exit: interactive TACTIC Beta and passing critical behavioral vectors in both adapters.

Task 6 — Integrated evaluation and Phase II design (Months 5–6). Run frozen experiment, ablations, matched manual comparison, failure taxonomy, ROI analysis, security/integration plan, and Phase II multi-asset design. Exit: Phase I Technical Validation Package and evidence-based Phase II go/no-go recommendation.

5.1 Milestones



5.2 Risk Reduction

- Reconstruction is photorealistic but not physics-ready — separate appearance from simulation geometry; use CAD/proxy colliders and gate on interaction fitness.
- Sparse/reflective imagery prevents reliable geometry — controlled capture, CAD fallback, and minimal-additional-view requests instead of hallucination.
- Articulation is plausible but wrong — evidence hierarchy, two-state/video observations, confidence gating, and mandatory SME validation.
- LLM invents a procedure/hazard — source-span requirement, constrained schema, deterministic validator, and 100% SME approval of safety-critical logic.
- Cross-modal alias maps to wrong part — alias/BOM/manual cross-reference, ranked candidates, and no-code correction.
- Engine semantics diverge — canonical Procedure IR, adapter-only engine specifics, and engine-independent conformance vectors.
- CUI cannot use public AI — local/private model abstraction, audit, SBOM/model inventory, and no mandatory public SaaS dependency.

5.3 Phase I Deliverables and Technical Data Products

The Base effort will produce six research artifacts in addition to the executable prototype. D1, Research & Test Protocol, documents requirements traceability, source-pack definitions, ground-truth labeling, metrics, baselines, security boundary, and software/model license register. D2, EvidencePack and Schema Specification, defines immutable evidence metadata, component/entity identifiers, SAG schema, provenance fields, and versioning rules. D3, Reconstruction and Articulation Bakeoff Report, records source configurations, geometry/appearance methods, component coverage, articulation results, failure cases, and the Month-3 geometry decision. D4, Semantic Compiler Alpha, contains the SAG/Procedure IR schemas, deterministic validator, provenance/confidence logic, and test vectors. D5, Interactive TACTIC Beta, contains the SME editor, primary runtime adapter, secondary-engine conformance path, and representative training package. D6, Phase I Technical Validation Package, reports the frozen experiment, manual comparison, H1–H5 results, ROI, failure taxonomy, cybersecurity/integration architecture, and evidence-based Phase II plan. This structure ensures the Navy receives reproducible technical evidence even if an individual algorithmic hypothesis fails.

5.4 Reproducibility, Software Supply Chain, and Model Governance

Every Phase I experiment will be reproducible from a versioned build manifest that records source hashes, parser/compiler versions, model identifiers, prompts/templates where applicable, container/environment versions, GPU/runtime configuration, and random seeds when supported. The project will maintain an SBOM plus a separate model/data-license inventory from Month 1. Candidate research code or models with restrictive, noncommercial, copyleft, data-use, or uncertain redistribution terms may be evaluated in an isolated research branch but will not silently become a product dependency. The production path will favor components that can be lawfully deployed in defense and commercial environments. Model selection is abstracted behind stable interfaces so a local approved model can replace a public/commercial service without changing the SAG or Procedure IR. These controls reduce supply-chain, licensing, reproducibility, and CUI-transition risk while preserving the ability to evaluate rapidly evolving 3-D and multimodal research.

6. Phase I Option Statement of Work — Six Months

The Option is a distinct six-month bridge to Phase II, not an unfocused continuation. O1: onboard a second electromechanical asset or materially different pump variant to test schema/ontology reuse and repeat key ablations. O2: improve collider generation, joint/limit inference, multi-part constraints, and reusable mechanical templates based on Base errors. O3: harden the secondary-engine/OpenXR deployment path, local inference, signed bundles, RBAC/audit, SBOM, and repeatable packaging. O4: define adaptive-tutor event APIs, multi-user/LVC adapter contracts, instructor scenario parameters, and a Navy-representative Phase II usability protocol. Exit evidence is a repeatable second-asset build, reduced correction burden, stronger cross-engine/deployment evidence, and a Phase II-ready interface/evaluation package.

7. Relationship to Phase II, Navy Transition, and Commercialization

Phase II will be a scale-and-generalize program. NV071 calls for diverse asset classes such as pumps and power supplies, automated kinematics/collision, adaptive tutoring and multi-user workflows, multiple export environments, Navy personnel usability assessment, Navy-relevant AR/MR deployment, and an IL-4/IL-5-oriented path [1]. The proposed sequence is: harden schema/versioning and asset onboarding; validate at least three materially different equipment classes; improve collision/kinematics/tool/fault models; productionize Unity/Unreal/OpenXR adapters; integrate tutor/multi-user/instructor APIs; harden local/private deployment and cybersecurity evidence; conduct Navy-representative authoring/usability studies following the required determination; and deliver SDK/API, installation, sustainment, and transition artifacts.

The Navy transition strategy is adapter-first: TACTIC-3D should feed existing/future immersive training environments rather than require replacement of the runtime ecosystem. Government/fleet SMEs validate source-grounded behavior once and publish to supported engines/devices. Phase III can integrate platform-specific technical data, LVC/training systems, sustainment/version-control workflows, and program-of-record deployment consistent with the topic's transition vision [1].

Commercial markets identified by the topic—commercial aerospace, advanced manufacturing, energy, and medical technology—face the same high-value equipment and procedural-content bottleneck [1]. TACTIC-3D will support secure self-hosted/private-cloud licensing for defense/regulated enterprises and enterprise SaaS/private tenants where policy permits. Revenue can combine platform subscription, asset-volume authoring, secure deployment, adapters, analytics, and services. TACTIC-3D will not claim novelty in XR training or no-code authoring. Its narrow differentiator is automated multimodal compilation: resolve technical entities to physical/3-D components, infer candidate articulation, compile procedures into deterministic source-grounded state logic, and export validated behavior through engine-neutral adapters.

7.1 Phase II Research Gates and Scaling Experiments

Phase II will use the Phase I corpus and variance estimates to define scaling gates rather than assuming one successful pump generalizes. Gate A, multi-asset semantics, will evaluate at least three materially different equipment classes—e.g., a centrifugal pump, an electrical/power-supply cabinet, and a more complex electromechanical assembly—using the same component/entity/procedure metrics. Gate B, kinematics and collision, will add automated collider simplification, compound joints, multi-body constraints, tool-object interactions, contact/clearance tests, and reusable mechanical templates, with SME acceptance and simulator conformance as the primary outcomes. Gate C, procedural richness, will add alternate branches, fault injection, diagnostic decisions, consumables, measurements, and parameterized scenarios while preserving deterministic validation and source provenance. Gate D, runtime and collaboration, will exercise production Unity/Unreal/OpenXR adapters, multi-user synchronization, instructor controls, telemetry, and an interface for adaptive tutoring/LVC systems. Gate E, secure deployment, will demonstrate containerized/local inference, role-based administration, vulnerability remediation, signed packages, audit export, backup/recovery, and an authorization-support evidence set for the Government-target environment.

Navy-representative usability testing will compare conventional authoring with TACTIC-assisted authoring on matched assets/procedures. Subject to the appropriate human-subjects determination, a within-subject crossover design will collect authoring labor, task completion, procedural defects, number/severity of SME corrections, workload/usability measures, and successful deployment of the authored lesson. Phase I timing and variance data will support a defensible Phase II sample-size calculation. The objective is not merely faster authoring; the system must preserve or improve procedural correctness and reviewer confidence while reducing specialized 3-D/XR labor.

7.2 Transition Architecture and Sustainment

Transition depends on keeping validated training semantics independent of any one engine, headset, or AI vendor. TACTIC packages therefore carry stable component IDs, versioned procedure/state logic, provenance, approval status, and conformance tests. Platform-specific adapters translate that package into engine-native rigid bodies, constraints, interaction affordances, cues, and telemetry. A change to a maintenance instruction can be recompiled and revalidated without rebuilding the entire 3-D lesson by hand; a change to the visualization/runtime layer does not invalidate the authoritative procedural source. Phase II will add package migration/versioning rules, backward-compatibility tests, signed release channels, reusable asset/procedure templates, and sustainment analytics that identify what changed between technical-publication revisions. This is central to the Phase III value proposition: reducing both initial content-production cost and the recurring cost of keeping immersive training synchronized with fleet technical data.

7.3 Commercial Positioning and Competitive Boundary

Commercial XR authoring platforms and simulation toolchains already validate demand for immersive maintenance and training; therefore TACTIC-3D will not position ordinary scene editing, headset delivery, or step-by-step authoring as novel. Its product boundary is the upstream compilation problem. A customer supplies engineering/visual evidence plus technical procedure data; TACTIC-3D produces a traceable behavioral digital twin whose components, articulation, allowable actions, states, faults, and procedure transitions are linked back to evidence and SME approvals. This allows the same validated semantic package to feed multiple runtime ecosystems. The business can sell secure platform licenses, per-asset compilation, enterprise adapters, private deployment, analytics, and transition/integration services while preserving the SAG/compiler/provenance/conformance layer as the durable product IP.

8. Related Work, Team Qualifications, Facilities, and Expected Results

8.1 Make/Buy Boundary and Foreground IP

TACTIC-3D integrates commodity/open components—CAD importers, photogrammetry, glTF/OpenUSD interchange, runtime engines, and replaceable foundation models—rather than reinventing them. SBIR R&D concentrates on the SAG ontology/evidence model, cross-modal entity resolution, articulation evidence fusion, source-grounded procedure extraction and deterministic Procedure IR compiler, confidence/provenance workflow, conformance suite, and no-code correction experience. This boundary reduces schedule risk and yields prime-owned technology that survives changes in individual models and 3-D representations.

8.2 Key Personnel and Workshare

Groundbreaker Solutions LLC will serve as prime and retain the semantic/AI/compiler architecture, integration, and majority of research effort. Principal Investigator Jason L. Lind will lead research design, architecture, integration, and Navy traceability. He brings more than two decades of software architecture experience across .NET/cloud systems, AI-enabled applications, cybersecurity-oriented architectures, simulation, mobile/web systems, and defense software modernization. The small-business team will combine roles as appropriate: AI/NLP engineer; 3-D/computer-vision engineer; training-domain SME; and a narrowly scoped Unity/Unreal/XR and reconstruction/articulation specialist subcontract. Final submission will name committed specialists. Core SAG, compiler, provenance, and model-routing code remains at the prime, and Base/Option workshare will remain safely above the DON two-thirds requirement [15].

8.3 Facilities

No unique Government-furnished laboratory is required for feasibility. Phase I uses controlled engineering workstations with modern GPU capability, selected XR hardware, smartphone/video capture equipment, a commercially obtainable or legally created representative test article, and secure source-control/build infrastructure. Local inference is supported so the research does not depend on external public-cloud AI. Government technical data/CUI, if furnished, will be handled only within the applicable approved contractor information-system boundary and contract controls.

8.4 Expected Phase I Results

Base completion delivers a functional prototype and quantitative evidence: frozen representative corpus and SME gold standard; secure multimodal ingestion; versioned SAG; component-to-geometry resolution; measured articulation hypotheses/no-code correction; source-grounded Procedure IR with deterministic validation; primary-engine interactive maintenance scenario and secondary-engine conformance proof; provenance/audit records; measured authoring, review, generation, test, and integration time versus manual baseline; failure taxonomy; and Phase II multi-asset/security/transition design. These outputs map directly to the topic's required end-to-end functional prototype, data parsing, low/no-code editing, Navy AR/MR integration planning, and ROI tracking [1].

8.5 SBIR Data Rights and Intellectual Property Strategy

Groundbreaker will maintain a background/foreground IP register from project start. Pre-existing company software, generic framework components, and third-party/open-source dependencies will be identified as background technology. New SAG ontology elements, evidence/provenance structures, cross-modal resolution methods, articulation-fusion logic, Procedure IR/compiler/validator, confidence-routing logic, conformance tests, and associated implementation developed under the effort will be tracked as foreground SBIR technology and marked/delivered in accordance with the awarded contract. Subcontract statements of work will be interface-bounded so specialists can implement runtime/reconstruction tasks without receiving ownership of the prime's compiler and semantic core. This supports future Phase III licensing and integration while protecting the Government's contractual rights and the small business's commercialization position.

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